

Name - Rutuja Waghamare

USN - 2VX24UE043

Expt.No- 3

Title - Merge Sort

```
In [2]: import random
import time
import matplotlib.pyplot as plt

def merge_sort(arr):
    if len(arr)<=1:
        return arr

    mid=len(arr)//2
    left=merge_sort(arr[:mid])
    right=merge_sort(arr[mid:])
    return merge(left,right)

def merge(left,right):
    result=[]
    i=j=0

    while i<len(left) and j<len(right):
        if left[i]<right[j]:
            result.append(left[i])
            i+=1
        else:
            result.append(right[j])
            j+=1
    return result+left[i:]+right[j:]

# input from user
```

```

arr=list(map(int,input("Enter elements separated by space:").split()))
print("Input array:",arr)

start_time=time.time()
sorted_arr=merge_sort(arr)
end_time=time.time()

print("Sorted array:",sorted_arr)
print("Time taken to sort:",end_time-start_time,"seconds")
print("=====")

# forn>5000
n_values=[5000,6000,7000,8000]
time_values=[]

for n in n_values:
    arr=[random.randint(1,9) for _ in range(n)]

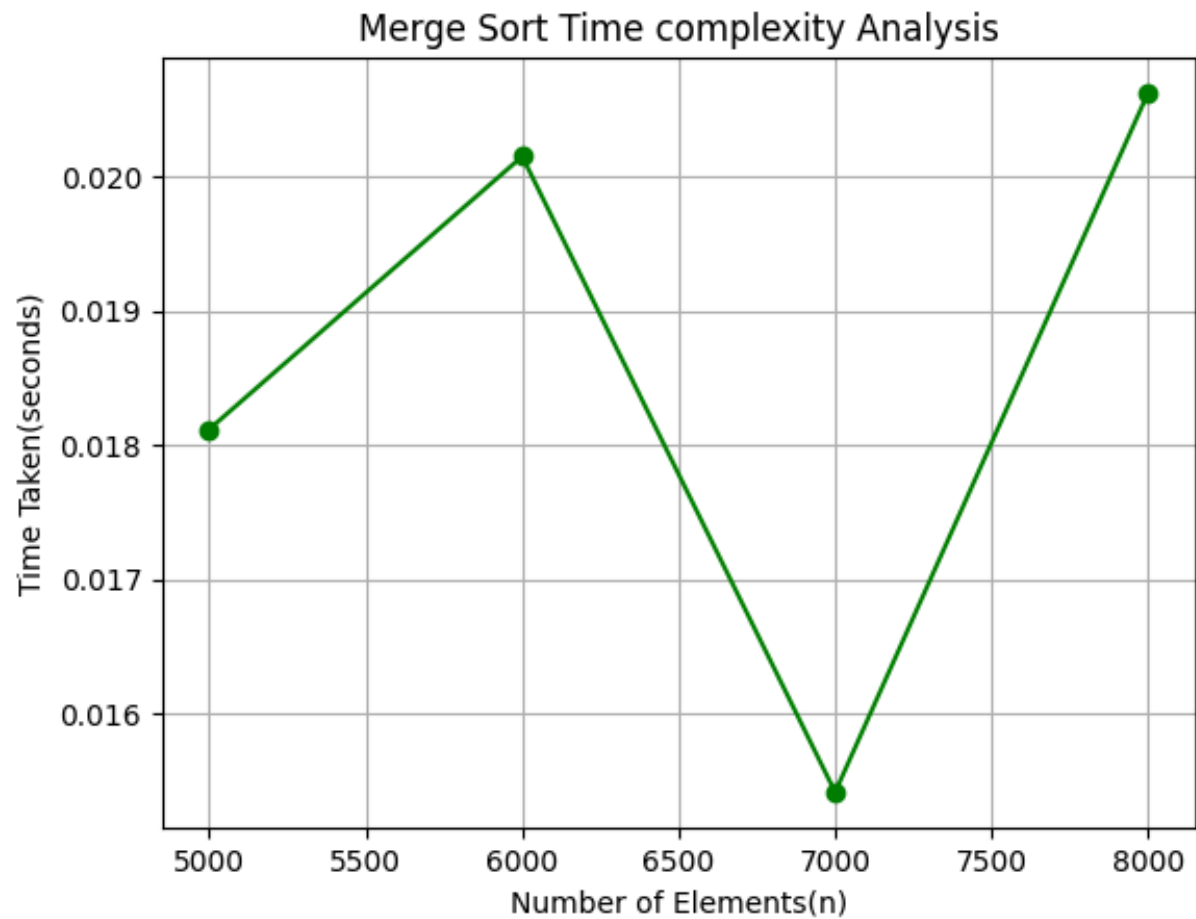
    start_time=time.time()
    merge_sort(arr)
    end_time=time.time()

    time_taken=end_time-start_time
    time_values.append(time_taken)
    print("Time for",n,"elements:",time_taken, "seconds")
print("=====")

# plotting the graph
plt.plot(n_values,time_values,color='green',marker='o')
plt.xlabel("Number of Elements(n)")
plt.ylabel("Time Taken(seconds)")
plt.title("Merge Sort Time complexity Analysis")
plt.grid()
plt.show()

```

```
Enter elements separated by space:5 3 4 2 1
Input array: [5, 3, 4, 2, 1]
Sorted array: [1, 2, 3, 4, 5]
Time taken to sort: 0.0 seconds
=====
Time for 5000 elements: 0.018118858337402344 seconds
Time for 6000 elements: 0.020153284072875977 seconds
Time for 7000 elements: 0.015412092208862305 seconds
Time for 8000 elements: 0.020627737045288086 seconds
=====
```



In []: